

CLAUDE PROJECT CONTEXT

AGENTIC AI-POWERED SCHOLARSHIP MATCHING SYSTEM

IMPORTANT CONTEXT

This document contains the complete project context.

Before proposing any architecture, code, workflow, UI, database structure or recommendation logic, please read and understand the entire project.

This project is being developed as part of:

PRESTIJ (Agentic AI for Educationists)

The project is currently at prototype stage.

The goal is NOT to build a production-ready national platform.

The goal is to demonstrate how Agentic AI can be used to provide personalised scholarship recommendations for Malaysian students after SPM.

PROJECT TITLE

Agentic AI-Powered Scholarship Matching System

PROJECT BACKGROUND

Many Malaysian students are unaware of scholarship opportunities that match their academic achievements, co-curricular involvement, interests and future aspirations.

Most students currently rely on:

Friends

Teachers

Counsellors

Social media

Scholarship websites

to discover scholarship opportunities.

The process is highly manual and students often:

Miss deadlines

Apply for unsuitable scholarships

Fail to discover relevant opportunities

Receive inconsistent guidance

The proposed solution is an Agentic AI-powered system that automatically analyses student profiles and recommends suitable scholarship opportunities.

PROBLEM DISCOVERY

Raw Pain Points

Scholarship Related

Students do not know what scholarship opportunities are available after SPM.

Students do not know which scholarships are suitable for them.

Students miss scholarship opportunities because they are unaware of application deadlines.

Students receive scholarship information from multiple websites and social media platforms.

Other Post-SPM Issues

Students do not know which university programme best matches their interests and strengths.

Students are uncertain about future career pathways.

Students receive conflicting advice from parents, teachers and peers.

Students do not fully understand their strengths and interests.

Students find it difficult to compare university, TVET and pre-university pathways.

Teachers and counsellors have limited time to provide personalised guidance.

CLUSTERING

Cluster A

Scholarship Awareness & Matching

Priority: TOP 1

Pain Points:

Lack of scholarship awareness

Difficulty identifying suitable scholarships

Missed scholarship deadlines

Scattered scholarship information

Root Cause:

No personalised scholarship matching system exists.

HMW:

How might we help students identify suitable scholarships based on academic results, co-curricular achievements, interests and aspirations?

PROJECT OBJECTIVE

Develop an Agentic AI-powered prototype that:

Analyses student profiles

Matches students with scholarships

Explains recommendations

Sends scholarship notifications

IMPORTANT PROJECT REQUIREMENTS

The project **MUST** remain scholarship-focused.

The project is **NOT**:

A university recommendation system

A career recommendation system

A counselling chatbot

These may support the process but the final output must always be:

SCHOLARSHIP RECOMMENDATION

TARGET USERS

Primary User:

SPM Graduates

Secondary Users:

School Counsellors

Teachers

Parents

Future Users:

DELIMa

Ministry of Education

DATASETS

The project will use Excel files.

No real database is required for the prototype.

Dataset 1

student_master.xlsx

Fields:

Student ID

Name

Email

Purpose:

Student identification

Login

Notification

Dataset 2

spm_results.xlsx

Fields:

Student ID

Subject Results

Academic Performance

Purpose:

Academic analysis

Status:

Not yet available.

May use simulated data.

Dataset 3

pajsk.xlsx

Fields:

GPA

CGPA

Merit

Leadership

Achievement

Participation

Status:

Available.

Dataset 4

psychometric.xlsx

Fields:

RIASEC

Career Interests

Personality Indicators

Status:

Not yet available.

May require support from school counsellors.

Dataset 5

student_aspiration.xlsx

Fields:

Dream Career

Preferred Field

Personal Interests

Important:

The project must not rely solely on historical student data.

Students should be allowed to declare their own interests.

Reason:

Students may participate in activities due to parental influence.

Observed behaviour may not reflect genuine interests.

Therefore self-declared aspirations are required.

Dataset 6

parent_background.xlsx

Fields:

Father Occupation

Mother Occupation

Parent Category

Categories:

B40

M40

T20

Important:

This dataset was suggested by the mentor.

Many scholarships consider socioeconomic background.

This dataset must be included.

Dataset 7

scholarship_database.xlsx

Fields:

Scholarship Name

Academic Requirement

PAJSK Requirement

Career Preference

Parent Category Requirement

Eligibility Criteria

Deadline

Status:

Currently under development.

AI AGENTS

The project should use SIX agents only.

Avoid creating too many agents.

Agent 1

Academic Profiling Agent

Role:

Analyse SPM results.

Output:

Academic profile.

Agent 2

PAJSK Analysis Agent

Role:

Analyse co-curricular achievements.

Output:

PAJSK profile.

Agent 3

Psychometric & Aspiration Agent

Role:

Analyse:

Psychometric data

Career interests

Student aspirations

Output:

Student interest profile.

Agent 4

Scholarship Matching Agent

Role:

Compare student profile with scholarship requirements.

Output:

Ranked scholarship recommendations.

Agent 5

Recommendation & Reasoning Agent

Role:

Explain recommendations.

Output:

Match percentage and recommendation reasoning.

Agent 6

Notification Agent

Role:

Send:

Scholarship alerts

Deadline reminders

Email notifications

SYSTEM WORKFLOW

Student Login ↓ Student Database ↓ SPM Results Dataset ↓ PAJSK Dataset ↓
Psychometric Dataset ↓ Student Aspiration Dataset ↓ Parent Background Dataset ↓
Academic Profiling Agent ↓ PAJSK Analysis Agent ↓ Psychometric & Aspiration Agent ↓
Scholarship Matching Agent ↓ Recommendation & Reasoning Agent ↓ Generate Top 5
Recommended Scholarships ↓ Notification Agent ↓ Email Alerts & Deadline Reminders
↓ Student Dashboard

USER INTERFACE REQUIREMENTS

IMPORTANT

The interface **MUST NOT** look like:

Cyberpunk

Gaming dashboard

Futuristic hacker UI

The interface **MUST** follow DELIMa design philosophy.

Requirements:

Educational platform appearance

White background

Blue and purple accents

Clean dashboard

Professional Ministry of Education style

Student-friendly layout

DEVICE SUPPORT

Must support:

Desktop

Mobile

Responsive design required.

LANGUAGE SUPPORT

Must support:

English

Bahasa Melayu

Language switching required.

TECHNICAL STACK

Prototype Development:

Google AI Studio (Gemini)

Backend:

Python

FastAPI

Frontend:

HTML

CSS

JavaScript

Data Storage:

Excel files

Server:

Uvicorn

EXPECTED OUTPUT

The system should generate:

Top 5 Recommended Scholarships

Match Percentage

Recommendation Explanation

Eligibility Analysis

Scholarship Deadline Reminder

Email Notification Status

FUTURE ENHANCEMENTS

Potential future integrations:

DELIMa

iDME

Real scholarship providers

Potential future datasets:

PBD

PBS

SEGAK

Potential future analytics:

Student development trend analysis

Longitudinal profiling from primary school to secondary school

These are future enhancements only and should not expand the scope of the current prototype.

FINAL INSTRUCTION FOR CLAUDE

Act as:

Agentic AI Architect

Educational Technology Solution Architect

Full Stack Developer

Scholarship Recommendation System Designer

Always prioritise simplicity, usability and prototype feasibility.

Do not over-engineer the solution.

The primary goal is to demonstrate a working Agentic AI scholarship matching prototype that could potentially be integrated into DELIMa in the future.